

HAFIZ MUHAMMAD

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PROFESSIONAL SUMMARY

AI & Backend Engineer specializing in building production-grade intelligent systems. Expert in architecting scalable **RAG pipelines** and **Autonomous AI Agents** using LangGraph and LangChain. Proficient in integrating multi-modal AI models with high-performance backends, secure payment gateways, and real-time data streaming architectures.

TECHNICAL SKILLS

AI & LLM Orchestration: LangChain, LangGraph, RAG (Retrieval-Augmented Generation), Agentic Workflows, Ollama, Groq, ChromaDB, BM25, all-MiniLM-L6-v2

Computer Vision & Audio: Whisper AI, PyannoteAI (Diarization), Wav2Lip, OpenCV, gTTS, Coqui TTS, Cartesia TTS (Low Latency)

Backend & Engineering: Python, FastAPI, Flask, Django, **Real-time Streaming**, RESTful APIs, Stripe API, Linux Process Management, FFmpeg

Database & Core: MySQL (Optimization), SQLAlchemy, NoSQL, Object-Oriented Programming (OOP), Data Structures (DSA), Git/GitHub

PROFESSIONAL EXPERIENCE

AI & Backend Engineering Intern

PureLogics, Lahore | July 2025 – Present

- Developing **real-time streaming** backend architectures for AI-driven automation using Python and FastAPI.
- Architecting modular codebases with **OOP principles** to integrate open-source LLMs (Ollama, Groq) into production workflows.
- Optimizing database query performance and Linux-level task management for high-concurrency AI processing.

AI-Powered Video Assistant (Internship Project)

PureLogics, Lahore | May 2025 – Aug 2025

- Built an end-to-end pipeline for automated transcription and **speaker identification** using Whisper and PyannoteAI.
- Implemented semantic search using **ChromaDB** and **all-MiniLM-L6-v2** embeddings for video content retrieval.
- Integrated FFmpeg for automated video/audio pre-processing within a Flask-based backend.

KEY PROJECTS

Movers – AI-Powered Vehicle & Logistics Platform

Architected a high-efficiency booking system featuring **Autonomous Dispatch Agents** built with **LangGraph**. Integrated **Real-time Tracking** and **Stripe API** for secure, automated payments. Developed a **RAG-based support layer** using **ChromaDB** and **all-MiniLM-L6-v2** to handle logistics queries, utilizing **Whisper** and **Cartesia TTS** for seamless voice coordination.

Tech: Flask, LangGraph, RAG, ChromaDB, Stripe API, Whisper, Cartesia TTS, Real-time Tracking, SQLAlchemy

Gourmet Haven AI (Cognitive Voice Agent)

Developed an autonomous voice agent using **LangGraph** for complex decision-making and **Cartesia TTS** for ultra-low latency real-time voice responses.

Tech: Python, LangGraph, RAG, Cartesia, Groq, Flask

AI Document Chat System (Multimodal)

Engineered a multimodal platform combining RAG with **Wav2Lip** and **Coqui TTS** to generate interactive AI-avatar document walkthroughs.

Tech: LangChain, ChromaDB, Wav2Lip, Whisper AI, MySQL, all-MiniLM-L6-v2

EDUCATION & CERTIFICATIONS

Bachelor of Science in Computer Science

Riphah International University, Lahore | Graduated Jan 2026

Core Competencies: AI & Machine Learning, Analysis of Algorithms, DBMS, Data Structures, Linux Management.

Certifications: Databases & SQL for Data Science, OOP Specialization, Design Patterns & Front-End.